

FILTER DESIGN TEST CASE

Low-Pass Filter - Band 66 (AWS) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting Band 66 (AWS) for IoT Narrowband Module. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 2136.7 MHz and bandwidth of 365.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: Band 66 (AWS)
- Application: IoT Narrowband Module
- Substrate: 128° YX LiNbO3
- Package: CSP (Chip Scale Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Andover HQ Lab

This specification applies to all variants of the Low-Pass Filter design targeting Band 66 (AWS). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2136.7 MHz
Bandwidth (3 dB):	365.6 MHz
Insertion Loss Target:	≤ 3.5 dB
Return Loss Target:	≥ 9.0 dB
Out-of-Band Rejection:	≥ 23.0 dB
Isolation (Duplexer):	≥ 28.0 dB
Q Factor:	3172
Temperature Coefficient:	-24.6 ppm/°C
Die Size:	1.1 x 1.6 mm
Wafer Size:	8-inch
Number of Resonators:	3
Electrode Material:	Mo

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Low-Pass Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 2136.7 MHz, BW 365.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 6410 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.5 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 548 MHz offset >= 23.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 6.6 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2024-10-10
Devices Tested: 11
Insertion Loss (meas): 3.25 dB
Return Loss (meas): 11.4 dB
Rejection (meas): 26.4 dB
Isolation (meas): 33.0 dB
Q Factor (meas): 3172
Group Delay Variation: 5.5 ns
Power Handling: +35 dBm
Die Yield: 86.5%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Low-Pass Filter design demonstrated excellent performance for Band 66 (AWS).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Group delay flatness improved compared to previous revision.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Reviewer:	Dr. Stefan Richter	Date:	2024-10-11
Approver:	Dr. Hans Weber	Date:	2024-10-22

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